

DATA ANALYTICS

PROFITABILITY & INVENTORY

2026

Retail Sales & Inventory Performance Dashboard

An analytical dashboard exploring the relationship between demand, profitability, discount strategy, and inventory efficiency.

Business Problem

High sales volume does not consistently translate into profitability.

Aggressive discounting drives demand but erodes margins, leading to inefficient inventory allocation and a significant share of unprofitable transactions despite overall growth.

Data source

Superstore — Tech, Office, Furniture

Analyst

Taif S.Alanazi

Tools

Python, Power BI

Pages

6

Executive overview

\$12.6M revenue · \$1.5M profit · but 25% of orders are unprofitable — growth is masking margin erosion


click for filters

Revenue

\$12.6M

Profit

\$1.5M

Profit margin %

11.6%

% unprofitable
orders
25%

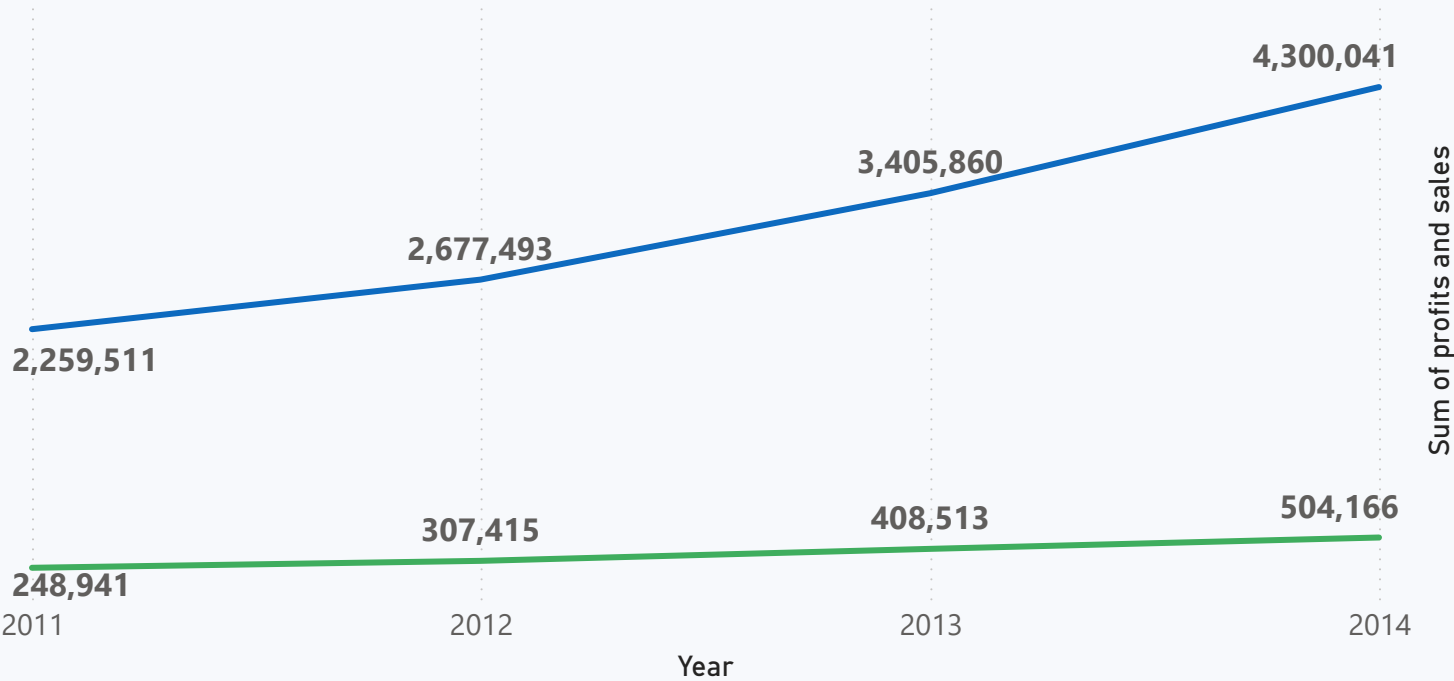
Quantity sold

178.3K

Sales and profits trend over years

Sales climbed from \$2.3M to \$4.3M over 4 years — but profit grew inconsistently

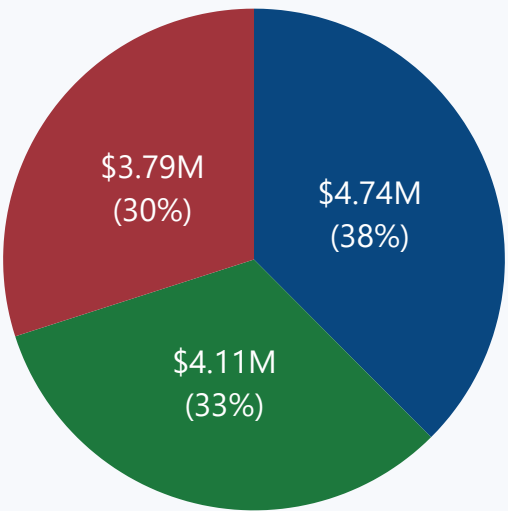
● Sum of profit ● Sum of sales



Sales by category

Technology Leads Revenue at 38%

● Technology ● Furniture ● Office Supplies



Discount Strategy Is Eroding Profitability

High discounts drive demand but destroy margins — discounts above 50% generate an average loss of \$98.89 per order

Discount impact

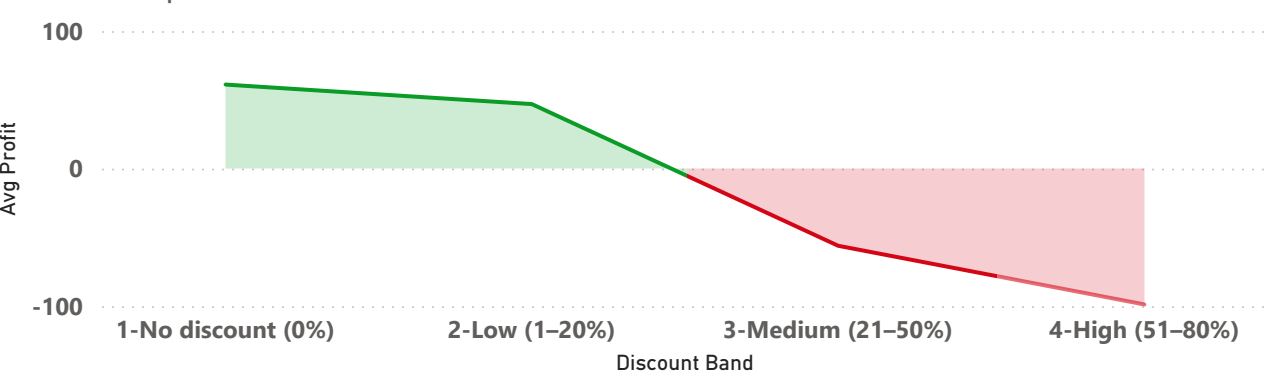
Avg profit of discount above 50%
-98.89

Discount–profit correlation
-0.32

% orders discount above 50%
9.46 %

Avg Profit by Discount Band

Profit Collapses After 50% Discount



Avg profit by discount level * sub-category

Every Sub-Category Loses Money Above 50% Discount

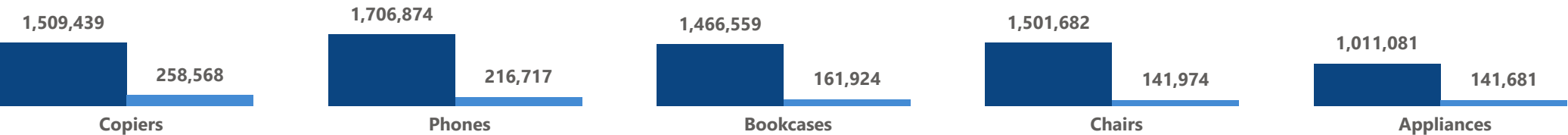
sub_category	1-No discount (0%)	2-Low (1-20%)	3-Medium (21-50%)	4-High (51-80%)
Accessories	75.63	33.92	-61.30	-75.49
Art	21.72	8.50	-17.13	-28.21
Binders	25.11	30.59	-16.03	-43.95
Chairs	114.16	52.49	-48.70	-173.93
Furnishings	35.49	14.78	-15.15	-50.25
Paper	25.00	16.24	-17.30	-24.66
Phones	142.23	63.75	-96.13	-207.86
Storage	50.40	27.14	-57.92	-73.40

Sales & profit

Sales vs Profit by sub category

High Sales Does Not Mean High Profit

● Sum of sales ● Sum of profit

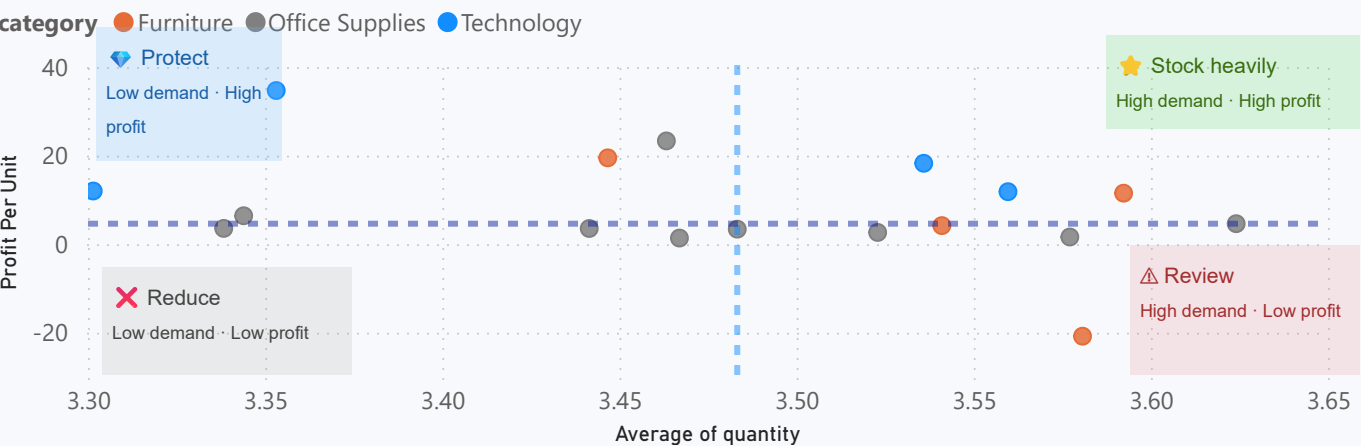


Inventory Strategy

Stock allocation analysis based on demand, profitability, and category policy

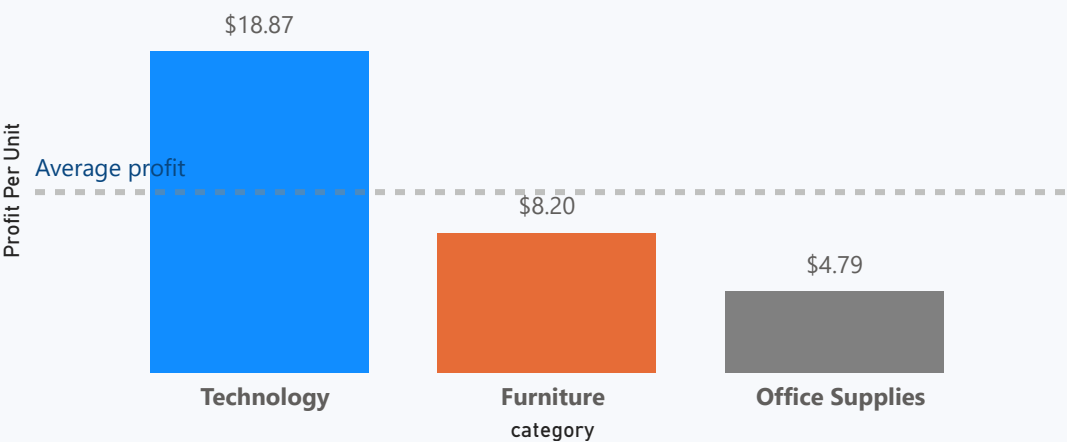
Inventory Risk Matrix

Products in bottom-right: high demand, low margin — review before restocking



Profit Per Unit by Category

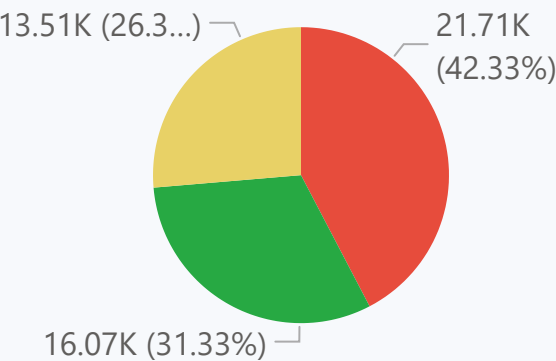
Technology earns 3.9× more per unit than Office Supplies



Stock Status Distribution

42% of orders are at reorder level — but profitability must be checked before restocking

Stock Status ● Reorder ● In stock ● Overstock



Products at Reorder Threshold

✓ Restock = profitable product · △ Review first = loss-maker — do not restock without pricing review

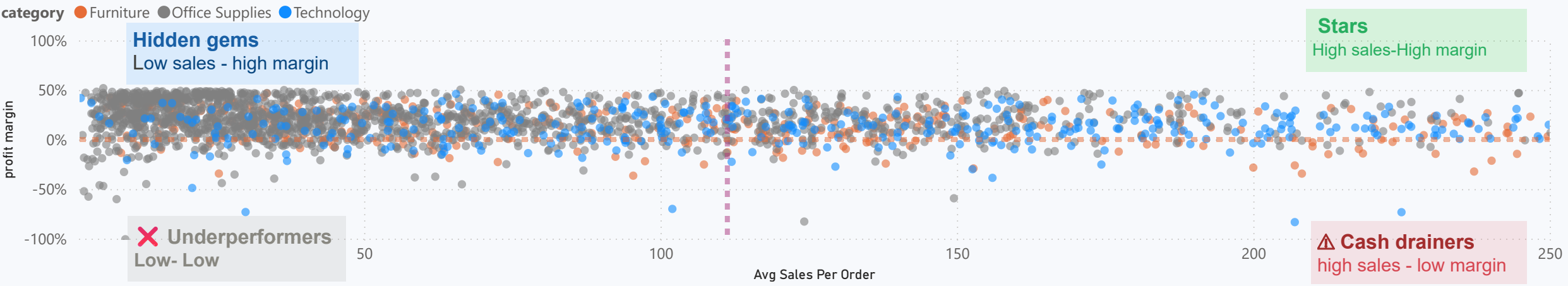
Product	category	Status	Avg profit	Stocking Action
12-1/2 Diameter Round Wall	Furniture	Reorder	10.39	✓ Restock
24-Hour Round Wall Clock	Furniture	Reorder	12.89	✓ Restock
36X48 HARDFLOOR CHAIRMAT	Furniture	Reorder	2.94	✓ Restock
36X48 HARDFLOOR CHAIRMAT	Furniture	Reorder	-13.85	△ Review first
3M Hangers With Command	Furniture	Reorder	1.91	✓ Restock
3M Hangers With Command	Furniture	Reorder	-1.41	△ Review first
3M Polarizing Light Filter	Furniture	Reorder	17.16	✓ Restock
3M Polarizing Task Lamp	Furniture	Reorder	71.23	✓ Restock
9-3/4 Diameter Round Wall	Furniture	Reorder	7.31	✓ Restock

Product Profitability

Evaluating product-level sales performance, margin efficiency, and profit sustainability

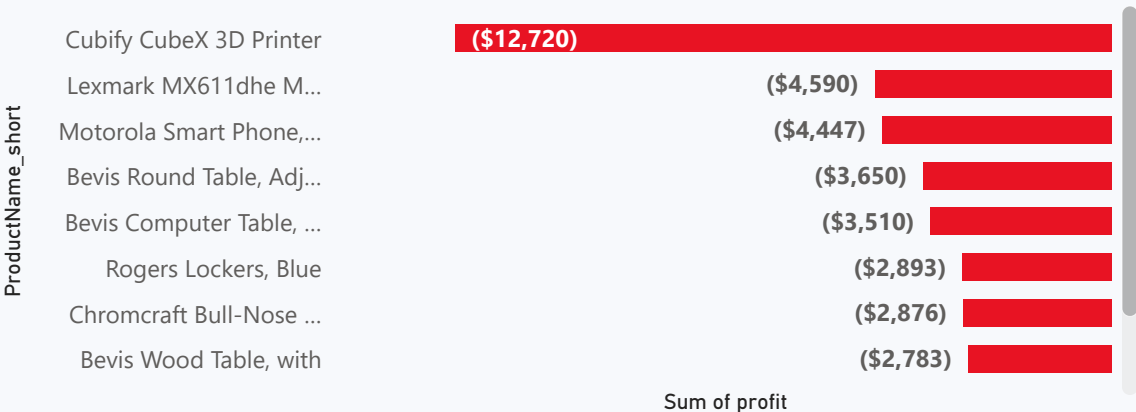
Product Profitability

Identifying which products to grow, fix, or discontinue based on sales and profit performance



Bottom 10 Products by Total Profit

These products are actively losing money — candidates for pricing review or discontinuation



Profit Trend by Product and Year

Are losses getting worse? Consistent red = discontinue · Recently red = adjust pricing. Total losses grew from −\$42K in 2011 to −\$81K in 2014 — nearly doubling in 4 years.

Product	2011	2012	2013	2014
Cubify CubeX 3D Printer		-2,639.99	-6,599.98	-3,479.99
Lexmark MX611dhe Monochrome Laser	-1,869.99			-2,719.98
Motorola Smart Phone, Cordless	218.53	252.46	-3,972.91	-945.11
Bevis Round Table, Adjustable	-780.64		-1,332.26	-1,536.99
Bevis Computer Table, Fully	-301.12	-563.80	-602.99	-2,041.64
Total	-42,741.19	-39,375.20	-63,277.03	-81,349.23

Strategic Recommendations

Four evidence-based actions derived from sales, discount, inventory, and product profitability analysis · 2011–2014

\$12.6M revenue · \$1.5M profit · 25% of orders unprofitable — profit margin flat at 11.6% for 4 years. Root cause: aggressive discounting and demand-driven inventory decisions eroding margins.

High Priority

Optimise Discount Strategy

- **Cap all discounts at 40%** — above 50% losses—\$98.89 loss per order on average.
- Target **Binders & Storage first** — highest 50–80% discount frequency in dataset.
- **Zero discounting on Technology** — Stars quadrant, highest margin category.
- Discounts above 50% appear in **9.46% of all orders** — immediate policy review needed.

Medium Priority

Prioritise High-Margin Products

- Redirect marketing toward **Phones & Copiers** — \$18.87 profit per unit vs \$4.79 Office Supplies.
- Technology generates **38% of revenue** yet consistently under-stocked vs Office Supplies.
- Canon imageCLASS: **\$25K+ profit from only 20 units** — highest profit-per-unit in dataset.
- Protect Technology margins — Phones lose **\$207 per order** when discounted above 50%..

Medium Priority

Align Inventory with Profitability

- Use **Inventory Risk Matrix** as primary stocking tool — high-demand products are not always high-margin.
- Check **Stocking Action** before every purchase order — ⚠ Review first means do not restock.
- **41% of orders at Reorder threshold** — profitability check required before restocking any product.
- Technology earns **3.9× more profit per unit** — stock allocation must reflect this advantage.

Ongoing

Review Loss-Making Products

- Total portfolio losses grew from –\$42K to –\$81K over 4 years — nearly doubled .
- **Red across all years** in profit trend matrix → discontinue — structural loss not fixable by discounting.
- **Recently turned red** → investigate discount promotions first — loss may be reversible.
- Profit grew from \$249K to \$504K but **sales grew faster** — track profit growth rate as monthly KPI.
- Review **Cash drainers quadrant** quarterly — high volume masks low per-unit returns.

Expected Impact — Grounded in Dataset Findings

Unprofitable Orders

25% → <10%

Page 2 · discount cap policy

High-Discount Avg Profit

–\$98.89 → >\$0

Page 3 · 40% cap enforcement

High-Discount Order Share

9.46% → <2%

Page 3 · orders above 50% discount

Strategic conclusion

Revenue grew from **\$2.3M to \$4.3M** over four years — yet profit margin remained flat at **11.6%** throughout. Commercial growth is not translating into financial efficiency.

The root cause is a **discount strategy that prioritises sales volume over margin quality**, compounded by inventory allocation that over-stocks low-margin Office Supplies while under-investing in high-margin Technology.

The path forward is not selling more — it is **selling smarter**: cap discounts, protect Technology margins, align every inventory decision with profit per unit, and discontinue products generating structural losses.